

Technical Document #650: Cybersecurity - Comprehensive Overview

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Penetration testing simulates cyber attacks to identify vulnerabilities. It helps organizations assess their security posture before real attacks occur.

Encryption converts plaintext into ciphertext to protect data confidentiality. AES and RSA are widely used encryption algorithms.

Security information and event management (SIEM) systems collect and analyze security logs. They provide real-time visibility into security events.

Intrusion detection systems (IDS) monitor network traffic for suspicious activity. They alert administrators of potential security breaches.

Network segmentation divides networks into smaller segments. It limits the spread of attacks and improves security monitoring.

Vulnerability management identifies, classifies, and remediates security vulnerabilities. Regular scanning and patching are essential components.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

Augmented reality overlays digital information on the physical world. It has applications in training, maintenance, and entertainment.

Sustainability and environmental responsibility are becoming key considerations in technology design. Green computing aims to reduce the environmental impact of technology.

Key Takeaways:

- Incident response plans define how organizations respond to security incidents.
- Security information and event management (SIEM) systems collect and analyze security logs.
- Multi-factor authentication (MFA) requires multiple forms of verification.